

A correction framework for day-ahead NWP solar irradiance forecast based on sparsely activated multivariate-shapelets information aggregation

Weijing Dou^a, Kai Wang^a, Shuo Shan^a, Chenxi Li^a, Kanjian Zhang^{a,b}, Haikun Wei^{a,b,*}, Victor Sreeram^c

^a Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, School of Automation, Southeast University, Nanjing 210096, China

^b Southeast University Shenzhen Research Institute, Shenzhen 518063, China

^c Department of Electrical, Electronics and Computer Engineering, The University of Western Australia, Perth, Australia

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ABSTRACT

Numerical Weather Prediction (NWP) is widely used in day-ahead solar irradiance forecast, which is of great significance to the optimization of power systems. Due to unescapable inherent errors of numerical techniques, NWP need correction. However, most correction schemes lack of error analysis, making the correction insufficiently efficient. Meanwhile, obtaining sufficient historical data in practical applications is challenging. Therefore, it is important to utilize the limited historical data to provide more meaningful information and increase the utilization of data information. To solve these problems, this paper proposes a day-ahead NWP solar irradiance correction framework. NWP global horizontal irradiance (GHI) error analysis is first conducted to determine the correction parts. Then, multivariate-shapelets analysis (MSA) is performed to select samples with high correlation to the correction parts. Mixture-of-experts (MoE) is adopted to sparsely activate high correlation samples contributing more to enhancing accuracy. Finally, a sequence-level information aggregation named SAIA is employed to obtain the corrected NWP forecasts. The proposed MSA-SAIA is evaluated with publicly available dataset and actual field dataset. The results demonstrate that MSA-SAIA yields the highest improvement, with increases of 16.15 % and 19.65 %. Additionally, we analyzed the performance of MSA-SAIA across different weather conditions, indicating its superior weather robustness.

1. Introduction

As one of the main renewable energy sources, solar energy plays an important role in meeting the requirements of global energy demand [1]. Adopting the inexhaustible and clean energy, solar energy, holds both immense potential and obstacles [2]. Recently, solar energy has been rapidly infiltrating the energy sector at an unprecedented pace, and this trend is expected to persist [3]. However, solar resources are intermittent and stochastic, which poses challenges for optimizing solar energy utilization and ensuring the grid's stability [4]. Accurate solar power forecast can further guarantee the related applications in the integrated power system [5,6], and solar power has a strong dependence on solar irradiance [7]. Therefore, accurately forecasting solar irradiance is particularly important for obtaining more accurate solar power forecasts [8].

Solar irradiance forecasts with different forecast horizons can meet various demands of grid operation, including ultra-short-term forecasts [9,10], long-term forecasts [11], medium-term forecasts [12,13],

and short-term forecasts [14,15]. The specific solar irradiance forecasting method should be determined based on the forecast horizon to ensure the most appropriate adoption [16]. Plenty of methods can be used for solar irradiance forecasting, such as persistence models, statistical models, and physical models, each suitable for different forecast horizons [17]. Numerical weather prediction (NWP), as a kind of physical model, is generally considered to be one of the best techniques for solar irradiance forecast beyond several hours (beyond 6 h [18]) ahead [19]. Especially when relatively detailed forecast scale information is required for longer forecast time horizons, NWP is more suitable [20]. The above advantages of NWP meet the requirements of day-ahead solar irradiance forecast tasks. For day-ahead forecast tasks, NWP is also regarded as an indispensable primary data source [21]. This is because forecast methods not based on NWP outcomes find it challenging to extend forecasts beyond several hours, resulting in insufficient guarantee of forecast accuracy [22].

* Corresponding author at: Key Laboratory of Measurement and Control of Complex Systems of Engineering, Ministry of Education, School of Automation, Southeast University, Nanjing 210096, China.

E-mail addresses: shuoshan@seu.edu.cn (S. Shan), hkwei@seu.edu.cn (H. Wei).

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Although NWP has shown relatively good performance in day-ahead solar irradiance forecasts, the errors still exist in NWP forecasts under certain circumstances due to uncertain initial conditions, unsimulated atmospheric phenomena, complex local topography, and the combined effects of numerical error propagation and approximations [23]. However, the forecast errors of NWP may be systematic, with patterns that typically exhibit learnable characteristics [24]. Hence, post-processing methods are commonly employed to correct the errors in NWP model outputs [25]. In previous studies, many post-processing methods have been developed to enhance the accuracy of NWP model outputs, such as machine learning and deep learning methods. Machine learning methods mainly include some regression models, containing linear regression [26,27], tree-based regression [28,29] and support vector regression (SVR) [30]. Common deep learning methods include long short-term memory (LSTM)-based models [31] and transformer-based models [32]. The literature indicates that the aforementioned post-processing methods can improve the accuracy of NWP solar irradiance forecasts. Nevertheless, these methods require a large amount of historical data to train correction models to ensure good correction accuracy.

In practical engineering applications, having a sufficiently large amount of historical data available for researchers to use in correcting NWP irradiance forecasts at specific field solar plants is challenging. Utilizing limited historical data to provide more meaningful information for the correction model is crucial for ensuring higher accuracy. To address this issue, the analog ensemble (AnEn) method has been proposed, primarily by searching a series of samples in the historical dataset that have high correlation with the correction target, thus providing the correction models with more useful and high-quality data information [33]. The AnEn-based methods have been widely used to improve the forecast accuracy of meteorological variables such as air quality, wind speed, and solar irradiance [34–38]. Zhou et al. developed a hybrid model based on similar day analysis (SDA), genetic algorithm (GA), and extreme learning machine (ELM) to predict hourly photovoltaic (PV) power outputs, where the day-level method SDA is used to select samples with high correlation to the target day for training the correction model [39]. The traditional AnEn searches for high correlation samples at the time-point level, while this SDA method extends the approach of establishing high correlation ensembles to day level. The above literature suggests that the approach based on establishing high correlation sets can provide more effective information for correction models, thus improving correction accuracy.

However, the methods mentioned above have the following limitations. Firstly, most existing correction methods [35,36] primarily correct the entire forecast outputs of NWP rather than selectively correcting specific parts (with relatively larger errors) based on the error analysis results, which leads to insufficient correction efficiency. The literature highlights the importance of evaluating the NWP forecast performance for guiding correction schemes [40], yet existing NWP irradiance correction schemes mostly lack an analysis of the errors in NWP forecasts. Secondly, existing solutions that search for high correlation samples at the day or time-point levels may result in low information utilization. For the correction segments selected based on analysis results, the methods [39] based on searching for a series of high correlation samples at the day level may introduce some redundant information. Meanwhile, the methods [33] that establish high correlation sets at the time-point level may result in the loss of some continuous information. Both of these approaches can potentially lead to poor correction accuracy. Additionally, existing approaches lack the screening of samples within the high correlation sets, which can lead to the inclusion of low-quality samples that have little positive impact or even a negative impact on improving correction accuracy.

To address the aforementioned issues and shortcomings, a hybrid correction framework for day-ahead NWP solar irradiance forecast is proposed based on sparsely activated multivariate-shapelets information aggregation. We first conduct NWP global horizontal irradiance

(GHI) error analysis in this work. Based on the results of this error analysis, the segments that require correction can be determined. The multivariate-shapelets analysis (MSA) [41] is performed on the correction parts to select a series of samples with high correlation to these segments, providing more meaningful data information to the correction model. Multivariate-shapelets is a novel form to search for high correlation samples, with the advantages of high accuracy and strong interpretability [42,43]. MSA, based on multivariate-shapelets, can balance the issues of data loss or redundancy, making it more effective in improving correction accuracy. Then, the mixture-of-experts (MoE) [44] layer is used to sparsely activate the high correlation samples searched by MSA. The MoE layer is a learnable module that can adaptively select samples that contribute more to improving the correction performance. Finally, a sequence-level information aggregation method [45] is used to aggregate the multivariate-shapelets information corresponding to the sparsely activated high correlation samples, obtaining the corrected NWP GHI forecasts. The performance and robustness of the proposed model named MSA-SAIA are evaluated based on quantitative metrics. The main contributions of this work are as follows:

- (1) NWP GHI error analysis is conducted to determine the segments requiring correction, thereby enhancing the targeting and efficiency of corrections. Performing corrections on segments with larger errors based on error analysis results is beneficial for preserving the generalization capability of the original NWP forecasts, and it also helps avoid potential overfitting issues that may arise from correcting the entire NWP outputs.
- (2) MSA is proposed to identify high correlation samples for each correction segment as inputs to the respective correction model. The advantages of using MSA to establish high correlation sets include avoiding data redundancy or loss and ensuring the quality of the selected high correlation samples.
- (3) MoE is adopted to sparsely activate the high correlation samples identified through MSA, allowing for the adaptive selection of samples that are more beneficial for improving correction accuracy. Introducing MoE contributes to ensuring that the correction scheme exhibits high accuracy and strong robustness.
- (4) The multivariate-shapelets information of sparsely activated high correlation samples are aggregated at sequence level, which can effectively enhance information utilization, thereby further improving correction performance.

The rest of the paper is arranged as follows: In Section 2, the materials and method are introduced including data description, NWP GHI analysis, and the details of MSA-SAIA. Using several evaluation indexes, the performance of the proposed correction model is compared with several commonly used correction models. Related results and discussions are presented in Section 3. Finally, the conclusions are drawn in Section 4.

2. Materials and method

In this section, the data used in this work is introduced. Then the NWP GHI error analysis results are given to select the parts that need to be corrected. Based on the above analysis results, this study proposes a day-ahead NWP GHI correction method, which mainly includes MSA and SAIA. The specific steps are as follows:

- (1) Data description is given in Section 2.1.
- (2) The overall procedure of the proposed correction approach is shown in Section 2.2.
- (3) NWP GHI error analysis results are shown in Section 2.3.
- (4) The details of MSA and SAIA in this paper are described in Sections 2.4 and 2.5.

Table 1
Details of the datasets used in this study.

Station	NWP model	Latitude	Longitude	Period	Temporal resolution	Meteorological information
AUS	ECMWF	23.76 S	133.88 E	01/01/2021–31/12/2022	1 (h)	GHI (W/m ²) AT (°C) RH (%)
YCZ	WRF	40.60 N	115.06 E	01/08/2018–31/07/2019	1 (h)	GHI (W/m ²) AT (°C) WS (m/s)

2.1. Data description

This study has two data sources: one is from a publicly available dataset (AUS) of a solar plant in Australia, and the other is from an actual field dataset (YCZ) of a photovoltaic power station in a province of China. The detailed information of AUS and YCZ are shown in Table 1. The meteorological information of AUS contains GHI, ambient temperature (AT), and relative humidity (RH). For YCZ, the meteorological information includes GHI, AT, and wind speed (WS). From Table 1, the period of AUS and YCZ are 24 months and 12 months, respectively. The temporal resolution of two studied sites is 1 h (24 points within 24 h). The NWP models for AUS and YCZ are sourced from the European Centre for Medium-Range Weather Forecasts (ECMWF) and the Weather Research and Forecasting (WRF), which was developed in the United States for mesoscale weather system emulation, respectively. The NWP GHI forecasts studied in this work are the 24-hour ahead forecast results issued daily at 0:00 (day-ahead).

2.2. The overall procedure of the proposed correction approach

The NWP GHI correction method proposed in this paper includes three stages, i.e. NWP GHI error analysis, MSA and SAIA. The flowchart is shown in Fig. 1. In the first stage, the main purpose of NWP GHI error analysis is to determine the part of NWP GHI prediction output that needs to be corrected. To achieve this goal, we use the observed GHI data and NWP GHI forecast data from the historical dataset to analyze the day-ahead forecast error of NWP GHI at each time point in detail. Since this work is oriented to the actual engineering needs, the most concerned index (NWP GHI error) in the actual assessment is selected as the evaluation index, which can better improve the correction efficiency.

In the second stage, a novel multivariate-shapelets analysis method, MSA, is proposed. Based on the NWP GHI error analysis, the length of the shapelets can be determined, equal to the length of the correction part. The advantage is that it can not only avoid the lack of input information, but also ensure that no redundant information is added. The multivariate-shapelets corresponding to the four meteorological factors included in the two sites are shown in Fig. 1. The multivariate-shapelets information used in the actual algorithm are as follows: NWP GHI, NWP AT, and NWP RH for AUS, and NWP GHI, NWP AT, and NWP WS for YCZ. The specific high correlation set of each correction part can be established by MSA. To further ensure correction accuracy, we utilize the sparse activation capability of MoE to screen samples in the high correlation set. Through MoE, the samples that contribute more to improving correction accuracy can be selected adaptively. In the final correction stage, we carry out the information aggregation (IA) based on the multivariate-shapelets of sparsely activated high correlation samples to obtain the final NWP GHI correction results. IA at the sequence level enhances utilization of data information, thereby improving correction accuracy.

2.3. NWP GHI error analysis

The GHI forecast values output by the NWP models are generally satisfactory and are widely used in various requirements of the actual power grid. Therefore, performing correction on periods with generally

large errors in NWP GHI predictions is more targeted and efficient. In practical engineering, it is necessary to correct the part where the NWP model cannot have a good forecast performance. There are many indicators to evaluate the performance of NWP, among which the NWP GHI error (deviation between observed values and forecasted values) is a metric of great concern in practical engineering applications. The reason is that the NWP GHI error reflects the magnitude of forecast errors and indicates whether the forecasts tend to overestimate or underestimate. NWP GHI error is calculated by the following equation:

$$error = f_i - o_i \quad (1)$$

where f_i and o_i are the NWP GHI and actual GHI observations at the moment i .

This study is aimed at day-ahead correction for the NWP GHI forecast. The forecast time horizon is 24 h (24 points). Therefore, the NWP GHI error at each point within 24 h is calculated and averaged to obtain the performance of NWP GHI forecasts at each time-point, as shown in Fig. 2. The part with poor forecast performance (exceeding the average of overall NWP GHI error) is selected for correction, and the error of the rest parts (the green range in Fig. 2) are relatively low. Hence these periods do not require correction. The overall average positive and negative errors are 62.12 W/m² and −80.07 W/m² for AUS, and 161.32 W/m² and −128.96 W/m² for YCZ, respectively. The selected correction parts comprise the union of periods exceeding the overall average positive and negative errors. Specifically, for this study, the part that needs to be corrected are from the 9th point to the 17th point (AUS) and from the 10th point to the 16th point (YCZ). For other sites, the specific correction part needs to be selected after the NWP GHI error analysis. It is worth noting that for other studies, the threshold for selecting correction parts can be adjusted according to specific practical application requirements.

2.4. Multivariate-shapelets analysis (MSA)

The purpose of MSA is to select a series of samples with high correlation to the correction part, and to group high correlation sets for training the correction model. Establish a high correlation set for each correction part, which can introduce more meaningful information to improve the correction accuracy and efficiency. Some common definitions in the MSA method can be found in [41]. For this study, the multivariate-shapelets referenced when searching for high correlation samples contain several factors, namely NWP GHI, NWP AT, NWP RH for AUS and NWP GHI, NWP AT, NWP WS for YCZ, respectively, as shown in Fig. 3. Adopting multivariate-shapelets can ensure that the data information is sufficient and reasonable, thereby enhancing the accuracy of identifying high correlation samples. The multivariate-shapelets of the correction parts are separately slid across their corresponding historical datasets, and the correlation is calculated respectively. The overall correlation D is as follows:

$$D = \sum_{i=1}^m w_i dist(A_{k,k+L}^i, S_i) \quad (2)$$

where $A_{k,k+L}^i$ represents the subsequence of length L of the i th factor in the historical dataset. S_i represents the i th shapelet. $dist$ represents the Euclidean distance. w_i represents the weight of the distance

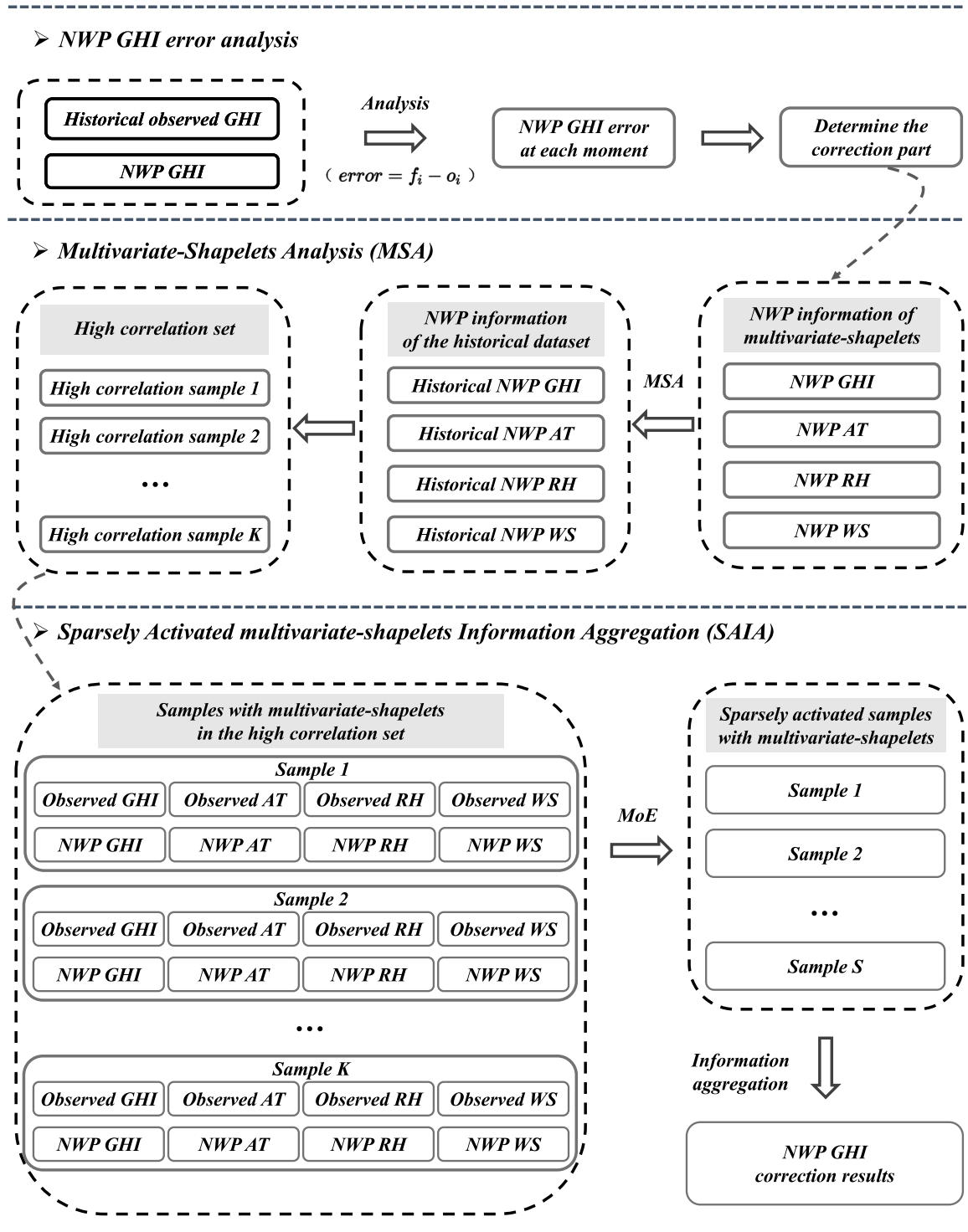


Fig. 1. Flow diagram of the proposed correction method. Note that the detailed data information for each step are listed.

corresponding to the i th factor, and w_i is determined by the Pearson's correlation coefficient between NWP GHI and the i th factor. m represents the number of factors.

Through the above MSA method, a series of samples with high correlation to the correction part can be selected to train the correction model. It should be noted that the multivariate-shapelets contain several factors. The overall correlation calculated in this way is the weighted result under the influence of several factors, resulting in

higher accuracy. And the length of the shapelets is equal to the length of the correction part, which is obtained by NWP GHI error analysis. Therefore, the information contained by multivariate-shapelets is sufficient and appropriate. For each correction part, high correlation sets are established targetedly, as illustrated in Fig. 3. For other works, it is suggested that the most suitable factors and corresponding weight w can be determined by the above method according to the actual datasets.

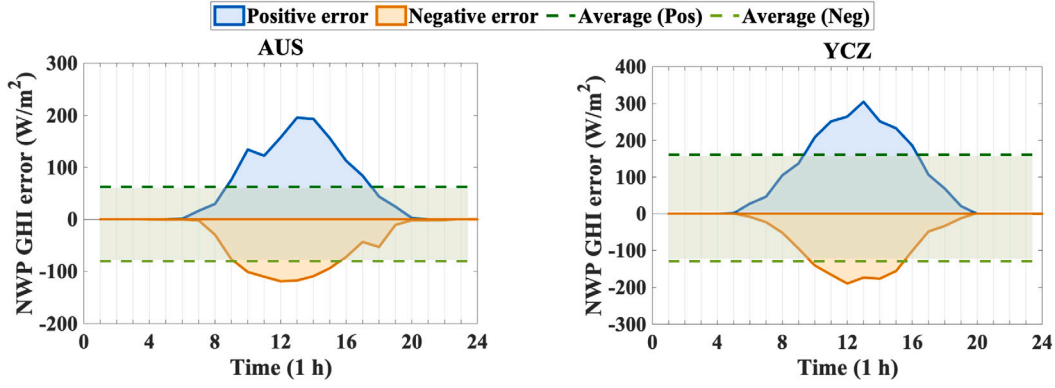


Fig. 2. NWP GHI analysis at each time-point within 24 h for (a) AUS and (b) YCZ.

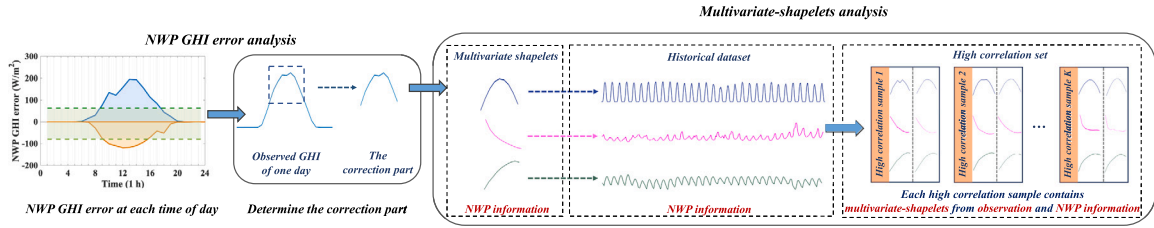


Fig. 3. The overview of multivariate-shapelets analysis (MSA).

2.5. Sparsely activated multivariate-shapelets information aggregation (SAIA)

This section introduces the correction scheme SAIA, which includes MoE and IA. The schematic diagram of the SAIA method is shown in Fig. 4. Based on the results of the aforementioned MSA, we established corresponding high correlation set for each correction part. We utilize MoE to filter samples within the high correlation sets to further ensure higher correction accuracy. MoE possesses the capability of sparse activation, which can be trained to select the samples that contribute more to improving correction accuracy. In this way, it facilitates the adaptive filtering of samples within the high correlation sets. Subsequently, the multivariate-shapelets information of the sparsely activated high correlation samples is aggregated to obtain the final correction results. It is worth noting that the IA is performed at the sequence level rather than point-level aggregation. This method significantly increases the utilization of data information, resulting in more accurate correction outcomes.

Mixture-of-experts (MoE). We embed a learnable MoE layer [44] into the correction module to adaptively select the samples in the high correlation sets. Through training, the high correlation samples, which contribute significantly to enhancing the accuracy of GHI forecasts, can be sparsely activated. Fig. 4 shows the internal structure of the MoE layer.

The MoE layer comprises two core components: a gating network G and a set of experts $E_1, \dots, E_M$. In this work, the input of G is the multivariate-shapelets of samples in high correlation sets. The output Y of the MoE layer is described as:

$$Y = \sum_{i=1}^M G(X)_i E_i(X) \quad (3)$$

where $G(X)$ the output of the gating network and $E_i(X)$ is the output of the i th expert network. M denotes the number of experts.

Notably, the gating network G demonstrates sparsity, enabling the MoE layer to selectively activate a few experts that are most advantageous for optimizing the results. This facilitates the adaptive selection

of samples within the high correlation set. Tunable Gaussian noise is introduced to the Softmax gating network to enhance load balancing. The top values are retained, while the remaining are set to $-\infty$. Specifically,

$$G(X) = \text{Softmax}(\text{KeepTopB}(H(X), b)) \quad (4)$$

$$H(X)_i = (X \cdot W_g)_i + \text{StandardNormal}() \cdot \text{Softplus}((X \cdot W_{\text{noise}})_i) \quad (5)$$

$$\text{KeepTopB}(v, b)_i = \begin{cases} v_i & \text{if } v_i \text{ is in the top } b \text{ elements of } v \\ -\infty & \text{otherwise.} \end{cases} \quad (6)$$

where W_g and W_{noise} are trainable weight matrix. W_{noise} is used to control the amount of noise per component.

Information aggregation (IA). As illustrated in Fig. 4, the IA [45] is carried out with the high correlation samples sparsely activated through MoE. For the multivariate-shapelets of the sparsely activated samples, we initially utilize multilayer perceptron (MLP) for dimension transformation, resulting in vectors $\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_S$ containing the data information for each sparsely activated sample. Then, the IA is conducted by the Softmax function. Directly aggregating the segments containing multivariate-shapelets information is more conducive to enhancing information utilization, thereby improving correction accuracy. The specific calculation equations are as follows:

$$\text{IA}(\mathcal{Z}) = \sum_{i=1}^S \mathcal{Z}_i \hat{\mathcal{Z}}_i \quad (7)$$

$$\hat{\mathcal{Z}}_1, \hat{\mathcal{Z}}_2, \dots, \hat{\mathcal{Z}}_S = \text{Softmax}(\mathcal{Z}_1, \mathcal{Z}_2, \dots, \mathcal{Z}_S),$$

$$\text{where } \hat{\mathcal{Z}}_i = \text{Softmax}(\mathcal{Z}_i) = \frac{\exp(\mathcal{Z}_i)}{\sum_S \exp(\mathcal{Z}_S)} \quad (8)$$

where $\mathcal{Z}$ denotes the information vector of multivariate-shapelets of the sparsely activated samples. S denotes the number of sparsely activated samples.

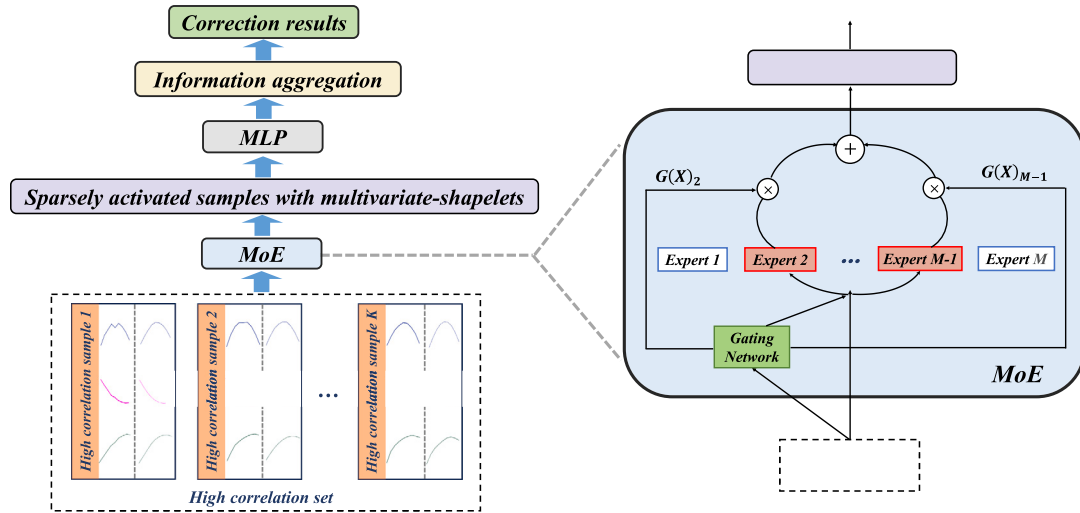


Fig. 4. The overview of sparsely activated multivariate-shapelets information aggregation (SAIA). A mixture of experts (MoE) layer is embedded for sparsely activating the samples that contribute more to improving the correction accuracy.

3. Results and discussion

To verify the reasonableness of the proposed method and its superiority compared with other methods, the analysis and discussion based on several commonly used evaluation criteria are given in this section.

3.1. Evaluation criteria and experimental setup

Besides NWP GHI error, root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), forecast skill (FS), and Pearson's correlation coefficient (R) are also calculated to verify the reasonableness of the proposed method and evaluate the correction results of these methods. These evaluation indexes are calculated as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (f_i - o_i)^2} \quad (9)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |f_i - o_i| \quad (10)$$

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{f_i - o_i}{o_i} \right| \times 100\% \quad (11)$$

$$\text{FS}(\%) = \left(1 - \frac{\text{RMSE}}{\text{RMSE}_{\text{NWP}}} \right) \times 100\% \quad (12)$$

$$\text{R}(\%) = \frac{\sum_{i=1}^n (o_i - \bar{o})(f_i - \bar{f})}{\sqrt{\sum_{i=1}^n (o_i - \bar{o})^2} \sqrt{\sum_{i=1}^n (f_i - \bar{f})^2}} \times 100\% \quad (13)$$

where f_i and o_i denote the NWP GHI and observed GHI at the moment i , respectively. n denotes total number of samples. $\bar{o}$ is the mean value of o_i , and $\bar{f}$ is the mean value of f_i .

The training, validation and testing set segmentation information for two sites is summarized as follows: for AUS, the training dataset covers the period from 11/2021 to 10/2022, with validation dataset from 11/2022 and the testing dataset from 12/2022. For YCZ, the training dataset spans from 03/2019 to 05/2019, the validation dataset covers 06/2019, and the testing dataset comprises data from 07/2019. In AUS and YCZ, the data preceding the correction target is used as the historical dataset to search high correlation samples. Furthermore, the temporal resolution of both AUS and YCZ is 1 h.

The proposed correction method comprises MSA and SAIA. Detailed descriptions of the input–output specifications and methodology for each module are provided in the preceding sections. The hyper-parameters for the proposed method are listed in Table 2.

Table 2

Hyper-parameters of the proposed method.

Module	Hyper-parameters	Value
MSA	Dimension of multivariate-shapelets	3
	Length of shapelets (AUS)	9
	Length of shapelets (YCZ)	7
SAIA	Input dimension (AUS)	(20, 9, 3)
	Input dimension (YCZ)	(20, 7, 3)
	Hidden size of MoE	128
	Number of experts in MoE	20
	Number of experts activated in MoE	5
	Number of hidden layers of MLP	1
	Hidden size of MLP	128
	Output dimension (AUS)	(9, 1)
	Output dimension (YCZ)	(7, 1)

3.2. Correction performance analysis

The core strategy of this study includes using MSA to establish high correlation sets and leveraging sparsely activated multivariate-shapelets for information aggregation at sequence level. The baseline models used in this study to validate the efficacy and efficiency of the proposed strategy are commonly employed approaches for NWP irradiance correction in the literature, including SVR [30], LSTM [46], temporal fusion transformer (TFT) [47], and AnEn [48]. AnEn is a traditional method in the literature, designed to improve correction performance by establishing high correlation sets. In AnEn, both the search for high correlation samples and the generation of correction results are conducted at the time-point level. MSA-general regression neural network (GRNN) [49] is an improved method compared to AnEn, which also achieves correction results at the time-point level. MSA-IA obtains correction results at the sequence level. Compared to the proposed method, MSA-IA lacks the sparse activation layer (MoE). The inclusion of MSA-IA as a comparison model aims to validate the effectiveness of information aggregation at the sequence level. The effectiveness of introducing MoE will be further analyzed in detail through ablation experiments in Section 3.4. The obtained correction results are analyzed through evaluation metrics as described in Section 3.1.

Table 3 shows the day-ahead NWP GHI correction performance of different models. Entries highlighted in bold indicate the best and second-best performances. In AUS and YCZ, SVR, LSTM, and TFT exhibit a relatively small improvement compared to the original NWP GHI forecasts. Regarding FS, the three models exhibit the maximum

Table 3
Day-ahead NWP GHI correction performance comparisons of various models.

Station	Models	Evaluation criteria				
		RMSE (W/m^2)	MAE (W/m^2)	MAPE (%)	FS (%)	R (%)
AUS	NWP	125.34	65.78	32.52	0.00	93.83
	SVR	120.24	63.21	31.17	3.99	94.21
	TFT	119.36	63.04	30.50	4.77	94.71
	LSTM	117.06	62.29	30.25	6.61	94.91
	AnEn	116.78	62.58	30.06	6.83	94.89
	MSA-GRNN	113.78	59.58	29.55	9.22	95.09
	MSA-IA	111.93	56.44	29.14	10.70	95.17
	MSA-SAIA	105.10	52.85	28.48	16.15	95.96
YCZ	NWP	209.93	111.57	170.75	0.00	84.43
	SVR	203.78	108.91	168.69	2.93	84.75
	TFT	194.27	106.78	156.68	7.46	84.92
	LSTM	197.70	107.63	159.79	5.83	84.89
	AnEn	185.95	103.80	157.70	11.42	85.27
	MSA-GRNN	183.97	103.31	151.38	12.37	85.03
	MSA-IA	175.52	103.53	141.06	16.39	85.18
	MSA-SAIA	168.67	99.61	136.67	19.65	86.97

improvements of 6.61% (AUS) and 7.46% (YCZ) respectively. This may be because such methods rely on a large amount of training data to achieve better correction accuracy. However, when historical data is relatively scarce, it may result in poor correction performance. The performance of AnEn outperforms that of SVR, TFT, and LSTM, as it enhances the correction accuracy by providing more relevant information through the construction of high correlation sets. However, the performance of AnEn is not as favorable as that of the MSA-based methods, as MSA utilizes multivariate-shapelets to select high correlation samples, effectively mitigating information loss or redundancy. Moreover, the performances of MSA-GRNN, MSA-IA, and MSA-SAIA are noticeably superior to other methods, indicating that the introduction of MSA can effectively enhance correction accuracy. This also indicates that when historical data is limited, training the correction model by searching for samples with high correlation to the correction target improves the correction accuracy. It helps to increase meaningful data information and alleviates the negative impact of insufficient training data.

For AUS and YCZ, the RMSE values for MSA-GRNN and MSA-IA are 113.78 W/m^2 and 111.93 W/m^2 , 183.97 W/m^2 and 175.52 W/m^2 ; the R values of these two models are 95.09% and 95.17%, 85.03% and 85.18%. MSA-IA outperforms MSA-GRNN, demonstrating the superiority of sequence-level aggregation. This is because aggregation at sequence level enhances the utilization of data information, thereby improving correction accuracy. For AUS and YCZ, the MAE values for MSA-IA and MSA-SAIA are 56.44 W/m^2 and 52.85 W/m^2 , 103.53 W/m^2 and 99.61 W/m^2 ; the MAPE values of these two models are 29.14% and 28.48%, 141.06% and 136.67%. This suggests that employing MoE to sparsely activate samples within high correlation sets provides advantages for enhancing the performance of the correction model. By introducing MoE, it can achieve the adaptive selection of high correlation samples that are beneficial for improving correction accuracy.

Fig. 5 illustrates the comparative forecast metrics of the models for forecast horizons of correction parts at 1-hour intervals. The performance of TFT and LSTM is notably inferior to that of the other methods, as the latter enhance the correction model by establishing high correlation sets, thereby providing more useful information from the limited data. The accuracy of the three MSA-based models exceeds that of AnEn, demonstrating the superiority of MSA. It can be seen from Fig. 5 that the proposed model achieves the minimum forecast error at almost all forecast horizons. Furthermore, for AUS and YCZ, the three methods introducing MSA exhibit relatively good correction accuracy across various forecast horizons. This indicates the superiority of establishing high correlation sets to increase more effective information. Two methods of sequence-level aggregation outperform

the point-level aggregation method, indicating that aggregating information at the sequence level allows for greater utilization of data information to enhance correction performance. The performance of MSA-SAIA is superior to MSA-IA, indicating the benefit of introducing MoE for adaptively selecting samples in high correlation sets.

3.3. Comparison under different weather conditions

This section presents a comprehensive analysis to assess the performance of the proposed method under different weather conditions. In this study, we primarily analyze the forecasting performance under three typical weather conditions that are commonly examined in previous research [50]: sunny, overcast, and cloudy. The daily clear sky index $K_{cs}(\text{day})$ [51] is used to define the daily weather conditions. Specifically, sunny conditions are characterized by $K_{cs}(\text{day}) > 0.9$, cloudy conditions by $0.5 < K_{cs}(\text{day}) < 0.9$, and overcast conditions by $K_{cs}(\text{day}) < 0.5$. Tables 4 and 5 show a comparison of day-ahead NWP GHI correction performance across various models under three weather conditions, with the bold results highlighting the best or second-best performance.

As shown in Tables 4 and 5, the performance of AnEn, MSA-GRNN, and MSA-SAIA is superior to that of LSTM and TFT. This is because these three methods, which are based on establishing high correlation sets, provide the correction models with more relevant information, thereby enhancing the correction accuracy. In general, the correction difficulty under sunny conditions is relatively low, and most models exhibit relatively good performance under sunny conditions. In contrast, under overcast and cloudy conditions, LSTM and TFT show poor correction performance, while AnEn, MSA-GRNN, and MSA-SAIA still demonstrate relatively good correction performance. This indicates that establishing corresponding high correlation sets for the correction samples helps improve the weather robustness of the correction model.

The accuracy of AnEn at the AUS site is slightly lower than that of MSA-GRNN and MSA-SAIA, but its performance at the YCZ site is significantly inferior to that of the other two methods. This is because MSA-GRNN and MSA-SAIA search high correlation samples based on multivariate-shapelets. This approach effectively balances the issues of information loss and redundancy, thereby further ensuring correction accuracy. For sunny, overcast, and cloudy, the proposed method consistently outperforms MSA-GRNN. This is attributed to the incorporation of the MoE module, which adaptively selects high correlation samples that contribute more to improving correction accuracy. The proposed method achieves the greatest improvement in NWP forecast correction under all three weather conditions, indicating that the proposed method has stronger weather robustness.

The error density analysis across three weather conditions is shown in Fig. 6, providing deeper insights into the distribution of masked

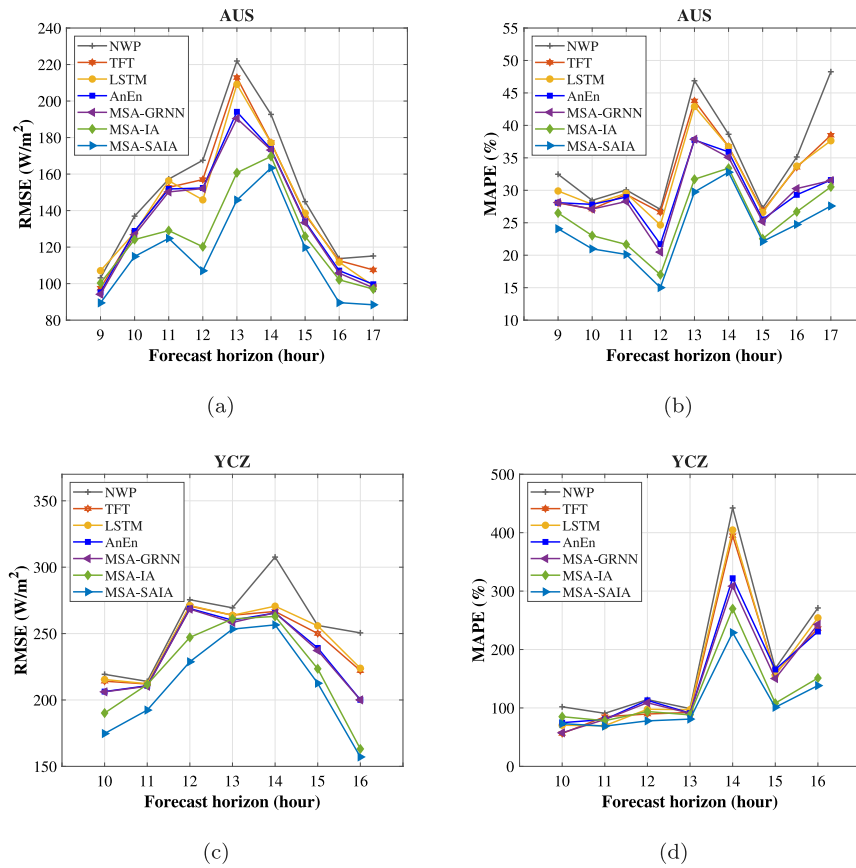


Fig. 5. The comparison results of day-ahead NWP GHI correction with different models.

Table 4

Day-ahead NWP GHI correction performance comparisons of various models under different weather conditions in AUS dataset.

Station	Weather conditions	Models	Evaluation criteria				
			RMSE (W/m ²)	MAE (W/m ²)	MAPE (%)	FS (%)	R (%)
AUS	Sunny [$K_{cs}(\text{day}) > 0.9$]	NWP	73.11	44.08	22.18	0.00	99.39
		TFT	53.72	30.58	20.98	26.52	99.41
		LSTM	60.40	35.83	21.39	17.38	99.50
		AnEn	46.54	24.98	20.58	36.34	99.53
		MSA-GRNN	45.20	26.66	20.77	38.18	99.49
		MSA-SAIA	43.82	24.63	20.52	40.06	99.54
	Overcast [$K_{cs}(\text{day}) < 0.5$]	NWP	149.34	79.96	42.78	0.00	79.05
		TFT	147.35	75.12	41.07	1.33	79.43
		LSTM	147.49	75.14	41.87	1.24	79.96
		AnEn	132.35	71.44	33.43	11.38	85.62
		MSA-GRNN	128.18	65.87	31.88	14.17	86.40
		MSA-SAIA	121.26	62.22	31.39	18.80	86.04
	Cloudy [$0.5 < K_{cs}(\text{day}) < 0.9$]	NWP	136.39	71.62	33.32	0.00	92.62
		TFT	135.72	71.54	33.26	0.49	92.72
		LSTM	133.66	70.87	33.15	2.00	93.42
		AnEn	133.32	70.20	33.14	2.25	92.78
		MSA-GRNN	130.53	69.12	32.96	4.30	93.24
		MSA-SAIA	121.75	65.64	31.93	10.73	93.90

errors. In most cases, the error distribution of the proposed MSA-SAIA exhibits a high density of residuals near 0 and a lower density of larger residuals (tails of the distribution). Notably, MSA-SAIA demonstrates clear superiority under all three weather conditions. Similar to previous results, TFT and LSTM exhibit inferior performance compared to the other three methods under all three weather conditions. As shown in Figs. 6(a) to 6(f), the performance of AnEn and MSA-GRNN is nearly on par with that of MSA-SAIA under overcast and cloudy conditions, but their performance is suboptimal under sunny conditions, particularly at the YCZ site. Additionally, MSA-SAIA consistently outperforms MSA-GRNN across all three weather conditions, which can be attributed

to the effective selection of high correlation samples facilitated by the MoE module. These results highlight the improved accuracy and weather robustness of NWP GHI correction achieved by integrating MSA with SAIA.

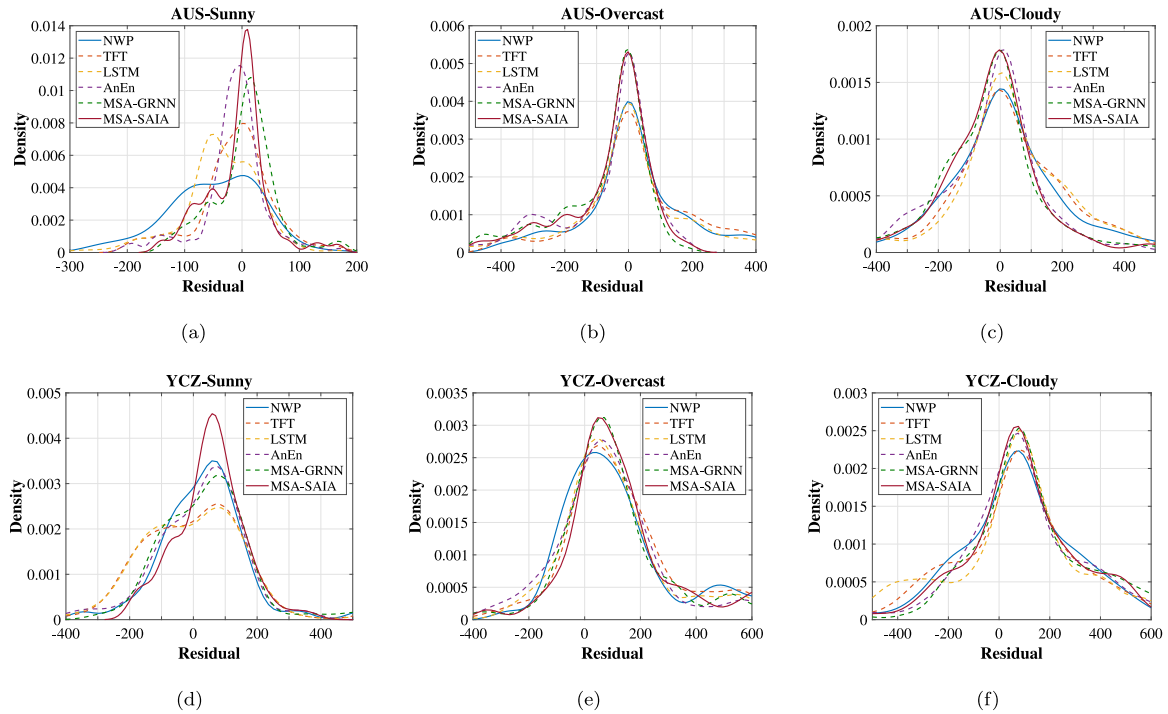
3.4. Sensitivity analysis

The correction performance of the proposed method under different training, validation, and testing set configurations is analyzed, as shown in Table 6. The results obtained by the configuration method introduced in Section 3.1 is labeled as Testing I. The original configuration

Table 5

Day-ahead NWP GHI correction performance comparisons of various models under different weather conditions in YCZ dataset.

Station	Weather conditions	Models	Evaluation criteria				
			RMSE (W/m^2)	MAE (W/m^2)	MAPE (%)	FS (%)	R (%)
YCZ	Sunny [$K_{cs}(\text{day}) > 0.9$]	NWP	137.14	65.71	38.29	0.00	94.02
		TFT	129.79	65.65	36.70	5.36	94.30
		LSTM	130.54	65.37	38.08	4.81	94.15
		AnEn	122.15	64.47	36.45	10.93	94.44
		MSA-GRNN	120.24	63.13	36.20	12.32	94.62
		MSA-SAIA	112.81	61.69	35.74	17.74	95.45
	Overcast [$K_{cs}(\text{day}) < 0.5$]	NWP	265.30	143.31	217.04	0.00	61.59
		TFT	226.66	121.85	197.76	14.56	63.22
		LSTM	228.98	126.73	199.65	13.69	64.74
		AnEn	210.65	120.63	195.79	20.60	68.19
		MSA-GRNN	184.29	101.64	179.77	30.54	65.47
		MSA-SAIA	171.68	92.81	172.71	35.29	68.53
	Cloudy [$0.5 < K_{cs}(\text{day}) < 0.9$]	NWP	218.34	120.15	193.70	0.00	82.92
		TFT	201.28	118.59	180.76	7.81	83.81
		LSTM	202.46	119.49	173.08	7.27	83.01
		AnEn	190.73	108.21	183.13	12.65	83.91
		MSA-GRNN	181.99	108.68	155.31	16.65	84.20
		MSA-SAIA	178.14	106.93	152.09	18.41	85.54

**Fig. 6.** Error distribution of correction models under different weather conditions in AUS and YCZ.

method is changed, specifically as follows: for AUS, the training dataset is from 11/2021 to 09/2022, the validation dataset is 10/2022, and the testing dataset is from 11/2022. For YCZ, the training dataset is from 02/2019 to 04/2019, the validation dataset is 05/2019, and the testing dataset is 06/2019. The results obtained by the following configuration method is labeled as Testing II. FS reflects the degree of improvement in forecast performance of the proposed method compared to NWP. As shown in Table 6, the proposed method achieves good correction performance under both Testing I and Testing II. This indicates that the proposed method is robust and generalizable across various configurations of data splits.

The optimal number of high correlation samples is determined by conducting experiments on the validation dataset, in which the number of samples in the training set (i.e., the high correlation set) is chosen differently. The validation datasets for AUS and YCZ are evenly divided into three subsets, designated as Validation I, Validation

II, and Validation III. The correction performance of the proposed model is analyzed on Validation I-III by varying the number of high correlation samples in the training set (10, 15, 20, and 25 samples). The corresponding experimental results are presented in the Table 7 below, where training (10), training (15), training (20) and training (25) represents the number of high correlation samples in the training set, specifically 10, 15, 20, and 25.

Table 7 presents the correction performance of the proposed method on Validation I-III when the number of high correlation samples in the training set is varied. For AUS, the performance of training (20) is generally better across Validation I-III. The average RMSE, MAE, and MAPE of training (20) are the lowest, with values of 100.47 W/m^2 , 52.12 W/m^2 , and 37.67%, respectively. Training (20) also exhibits the highest average R, which is 96.25%. For YCZ, training (20) achieves the lowest average RMSE and MAPE, along with the highest R. However, the average MAE of training (20) is not the lowest, being 1.93 W/m^2 .

Table 6
Day-ahead NWP GHI correction results obtained using different training-validation-testing set configurations.

Station	Testing set	Models	Evaluation criteria				
			RMSE (W/m^2)	MAE (W/m^2)	MAPE (%)	FS (%)	R (%)
AUS	Testing I	NWP	125.34	65.78	32.52	0.00	93.83
		MSA-SAIA	105.10	52.85	28.48	16.15	95.96
	Testing II	NWP	115.08	57.57	32.81	0.00	95.44
		MSA-SAIA	102.51	52.30	24.99	10.92	96.26
YCZ	Testing I	NWP	209.93	111.57	170.75	0.00	84.43
		MSA-SAIA	168.67	99.61	136.67	19.65	86.97
	Testing II	NWP	183.47	99.49	96.59	0.00	89.70
		MSA-SAIA	150.10	83.03	89.03	18.19	90.11

Table 7
Day-ahead NWP GHI correction results obtained by selecting different numbers of high correlation samples as the training sets.

Station	Validation set	Evaluation criteria	Training set			
			Training (10)	Training (15)	Training (20)	Training (25)
AUS	Validation I	RMSE (W/m^2)	84.88	86.04	85.45	85.57
		MAE (W/m^2)	46.59	46.58	46.65	45.82
		MAPE (%)	29.69	30.91	28.73	31.88
		R (%)	98.08	98.12	98.03	98.06
	Validation II	RMSE (W/m^2)	117.77	106.85	110.80	109.13
		MAE (W/m^2)	59.60	54.98	56.83	55.32
		MAPE (%)	45.56	44.83	45.01	46.78
		R (%)	96.01	96.30	96.23	96.22
	Validation III	RMSE (W/m^2)	110.86	110.21	105.15	114.14
		MAE (W/m^2)	57.07	54.89	52.87	56.93
		MAPE (%)	37.78	43.77	39.27	43.36
		R (%)	93.95	94.19	94.48	93.70
	Average	RMSE (W/m^2)	104.50	101.03	100.47	102.95
		MAE (W/m^2)	54.42	52.15	52.12	52.69
		MAPE (%)	37.68	39.83	37.67	40.67
		R (%)	96.02	96.20	96.25	95.99
YCZ	Validation I	RMSE (W/m^2)	234.26	216.89	209.62	207.15
		MAE (W/m^2)	144.92	143.65	143.76	145.96
		MAPE (%)	95.74	91.87	88.53	96.97
		R (%)	81.62	81.49	81.71	81.62
	Validation II	RMSE (W/m^2)	163.85	164.90	167.06	172.95
		MAE (W/m^2)	102.48	116.73	124.36	130.96
		MAPE (%)	332.74	303.59	295.34	311.08
		R (%)	89.77	89.81	89.95	89.56
	Validation III	RMSE (W/m^2)	227.30	203.12	197.36	199.51
		MAE (W/m^2)	156.82	144.31	141.89	144.26
		MAPE (%)	68.36	61.02	58.58	70.33
		R (%)	80.71	80.82	80.90	79.96
	Average	RMSE (W/m^2)	208.47	194.97	191.35	193.20
		MAE (W/m^2)	134.74	134.90	136.67	140.39
		MAPE (%)	165.61	152.16	147.48	159.46
		R (%)	84.03	84.04	84.19	83.71

higher than the minimum average MAE. Considering four evaluation metrics, training (20) yields the best performance. Therefore, for both AUS and YCZ, the number of high correlation samples in the training set is set to 20.

3.5. Ablation study

The ablation study aims to verify the effectiveness of each component of the proposed correction method, specifically MSA and SAIA. This day-ahead NWP GHI correction method is designed for 24-hour ahead NWP GHI forecasts, covering the period from 0:00 to 24:00. Based on the results of NWP GHI error analysis, the correction parts for AUS and YCZ are selected, while the remaining parts retain the original forecast results of the NWP models.

Table 8 shows the results of the ablation study, in which the results marked in bold black indicate the best correction performance. The correction results obtained by MSA-IA are better compared with NWP and IA. This validates the effectiveness of improving the accuracy of NWP GHI forecasts by searching for a series of samples with high

correlation to the target within the historical dataset and aggregating their multivariate-shapelets information. The advantage of MSA is that it can make the high correlation sets more accurate and interpretable. Introducing MoE, the correction accuracy by MSA-SAIA is greatly improved compared with the results of MSA-IA. It means that MoE is effective in improving the correction accuracy, as MoE can filter out some low-quality samples. Therefore, the proposed scheme is beneficial to improve the day-ahead NWP GHI correction accuracy.

Fig. 7 displays the real GHI series (gray lines), the NWP GHI series (blue lines), and the corrected GHI series from various models, which shows the correction performance of different models under various sky conditions. From the GHI curves of Fig. 7, it can be seen that the MSA-SAIA and MSA-IA yield relatively good correction results, making the NWP GHI forecasts more closely with the observed GHI curves. However, these correction models cannot be very sensitive to sudden changes of GHI. The correction result curves of MSA-SAIA are closer to the observed GHI curves in most cases. Overall, it is more robust for the MSA-SAIA to achieve better performance, since the MoE can

Table 8
Ablation results of the proposed approach for day-ahead NWP GHI correction performance.

Station	Models	Evaluation criteria				
		RMSE (W/m^2)	MAE (W/m^2)	MAPE (%)	FS (%)	R (%)
AUS	NWP	125.34	65.78	32.52	0.00	93.83
	IA	119.10	62.90	30.94	4.98	94.51
	MSA-IA	111.93	56.44	29.14	10.70	95.17
	MSA-SAIA	105.10	52.85	28.48	16.15	95.96
YCZ	NWP	209.93	111.57	170.75	0.00	84.43
	IA	198.67	107.31	166.06	5.36	84.87
	MSA-IA	175.52	103.53	141.06	16.39	85.18
	MSA-SAIA	168.67	99.61	136.67	19.65	86.97

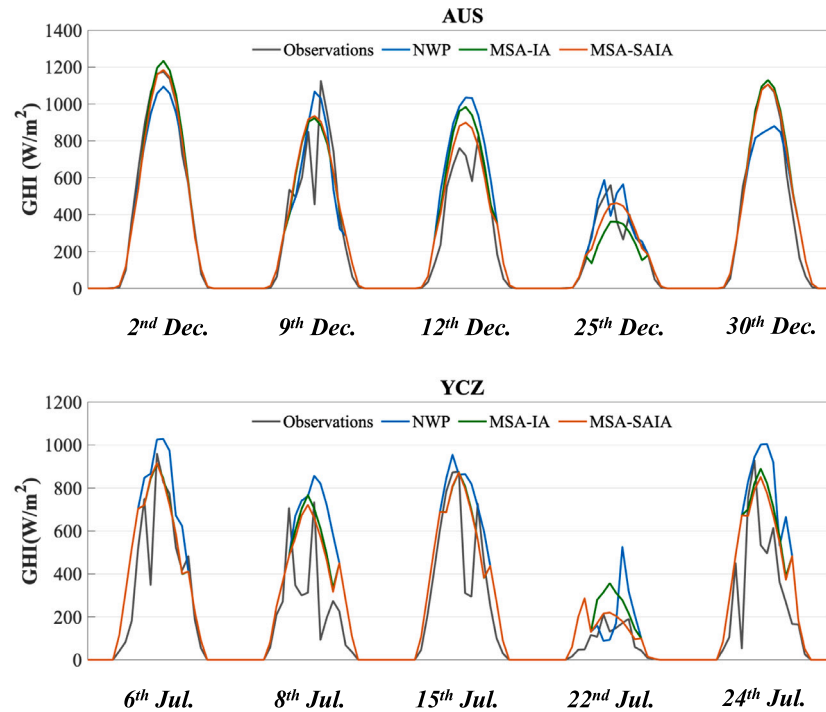


Fig. 7. The GHI correction results of different models in AUS and YCZ.

sparsely active the samples that make more contribution to improving correction accuracy under different conditions.

3.6. Further exploration

This section discusses the computational cost of the model to explore the practicality of the proposed method. We discuss the computational cost of the models using the YCZ dataset as an example. The inference time for correcting a single sample is recorded for each model and module over 100 trials, and the average results are summarized in the Table 9. The proposed method consists of three modules: MSA, SA, and IA, with corresponding inference time of 319.09 ms, 15.52 ms, and 0.7 ms, respectively. The incorporation of the MSA and SA modules notably improves correction accuracy. The additional inference time introduced by these two modules does not have a noticeable impact on the day-ahead correction task.

As shown in Table 9, the inference times for AnEn, MSA-GRNN, and MSA-SAIA are 320.22 ms, 329.19 ms, and 335.91 ms, respectively, which are higher than those of the SVR, TFT, and LSTM. This is due to the additional inference time required for searching high correlation samples within the historical dataset. Although the proposed MSA-SAIA model exhibits the largest inference time, it is far less than the forecasting horizon for the day-ahead correction task, which does not impact the practical applicability. For practical correction tasks, the accuracy

Table 9

Computational cost of the proposed method and baseline methods. An example of the experiment with day-ahead correction in YCZ.

Model	Inference time	Model	Inference time
SVR	4.85 ms	MSA	319.69 ms
LSTM	12.14 ms	SA	15.52 ms
TFT	52.32 ms	IA	0.70 ms
AnEn	320.22 ms	SAIA	16.22 ms
MSA-GRNN	329.19 ms	MSA-SAIA	335.91 ms

of the correction model is of greater concern. The proposed MSA-SAIA method achieves high correction accuracy, providing valuable insights for tasks such as grid scheduling and optimal utilization of PV power.

In Section 2.3, NWP GHI error analysis is performed to determine the correction parts. Here we also analyze the correction performance of MSA-SAIA for the selected correction part, as shown in Fig. 8. Figs. 8(a)–8(d) represent the positive and negative errors of the NWP and MSA-SAIA at each time of correction part for AUS and YCZ, aiming to show the detailed correction performance. The height and color intensity of the bars in each subplot depict the magnitude of the corresponding NWP GHI error. In most cases, the positive and negative errors of MSA-SAIA are smaller than the corresponding errors of NWP, indicating that the proposed method has a significant correction effect at almost all time of the correction parts. However, at certain times, the

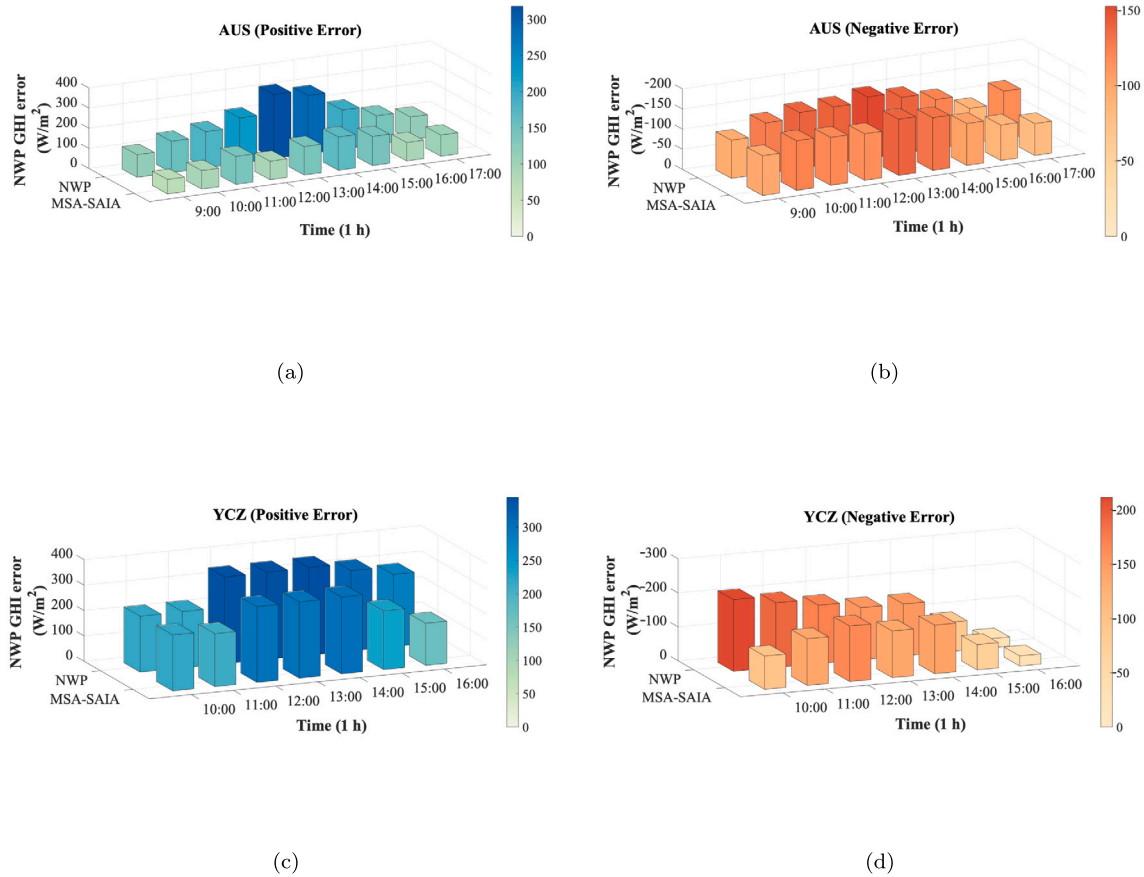


Fig. 8. Average of NWP GHI error at each time of correction part in AUS and YCZ.

positive and negative errors of MSA-SAIA are close to the corresponding errors of NWP, particularly at YCZ, where this case is more obvious. This could be attributed to the relatively limited historical dataset available for the actual field dataset YCZ site, which may not encompass a sufficient variety of high correlation samples, thereby resulting in less pronounced correction effects. As the historical dataset accumulates, the proposed method can search for high correlation samples that better match the target, potentially leading to further improvements in correction accuracy.

4. Conclusions

Accurate solar irradiance forecast significantly influences PV power forecast, thereby impacting the dispatching and operation of power systems. This paper has proposed a novel scheme for day-ahead NWP GHI correction. The NWP GHI error analysis is first carried out to determine the correction parts, aiming to enhance the efficiency of the correction process. MSA is performed to select a series of samples that have high correlation to the correction parts and high correlation sets are established with these samples to train the correction model. It is worth mentioning that the respective high correlation sets are established for each correction part, which can introduce more meaningful data information. MSA is based on multivariate-shapelets, with the advantages of high accuracy and strong interpretability. For the high correlation samples searched by MSA, MoE is utilized to sparsely activate the samples that contribute more significantly to improving correction accuracy. Note that MoE is a learnable module that can achieve adaptive selection of samples in high correlation sets. Finally, to further improve information utilization, a sequence-level aggregation method SAIA is employed to obtain the NWP GHI correction results.

The effectiveness of the proposed scheme has been demonstrated by experiments with both publicly available dataset and actual field dataset. The results have verified the outstanding ability of the proposed scheme to improve the day-ahead correction accuracy compared with other methods. For AUS and YCZ, MSA-SAIA shows the highest improvement for NWP forecasts, with increases of 16.15% and 19.65%. Moreover, the correction performance of MSA-SAIA under different weather conditions are also discussed. A series of results suggest that our proposed method demonstrates stronger weather robustness. It is worth noting that we conduct a comprehensive analysis of the correction performance at each time of correction parts. The results indicate that MSA-SAIA can improve the accuracy of NWP GHI forecasts at most time of correction parts.

The proposed method has the benefit of making solar power forecasts more accurate, which is of great significance for ensuring the stable operation of the grid. Furthermore, it is equally important for wind power forecasts to correct NWP wind speed accurately. In fact, solar energy and wind energy are complementary in terms of renewable power generation. In future work, this proposed model is planned to be adjusted to correct the NWP wind speed.

CRedit authorship contribution statement

Weijing Dou: Writing – review & editing, Writing – original draft, Software, Methodology, Formal analysis. **Kai Wang:** Writing – review & editing, Software, Methodology. **Shuo Shan:** Writing – review & editing, Supervision, Methodology. **Chenxi Li:** Writing – review & editing, Supervision. **Kanjian Zhang:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Haikun Wei:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition. **Victor Sreeram:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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